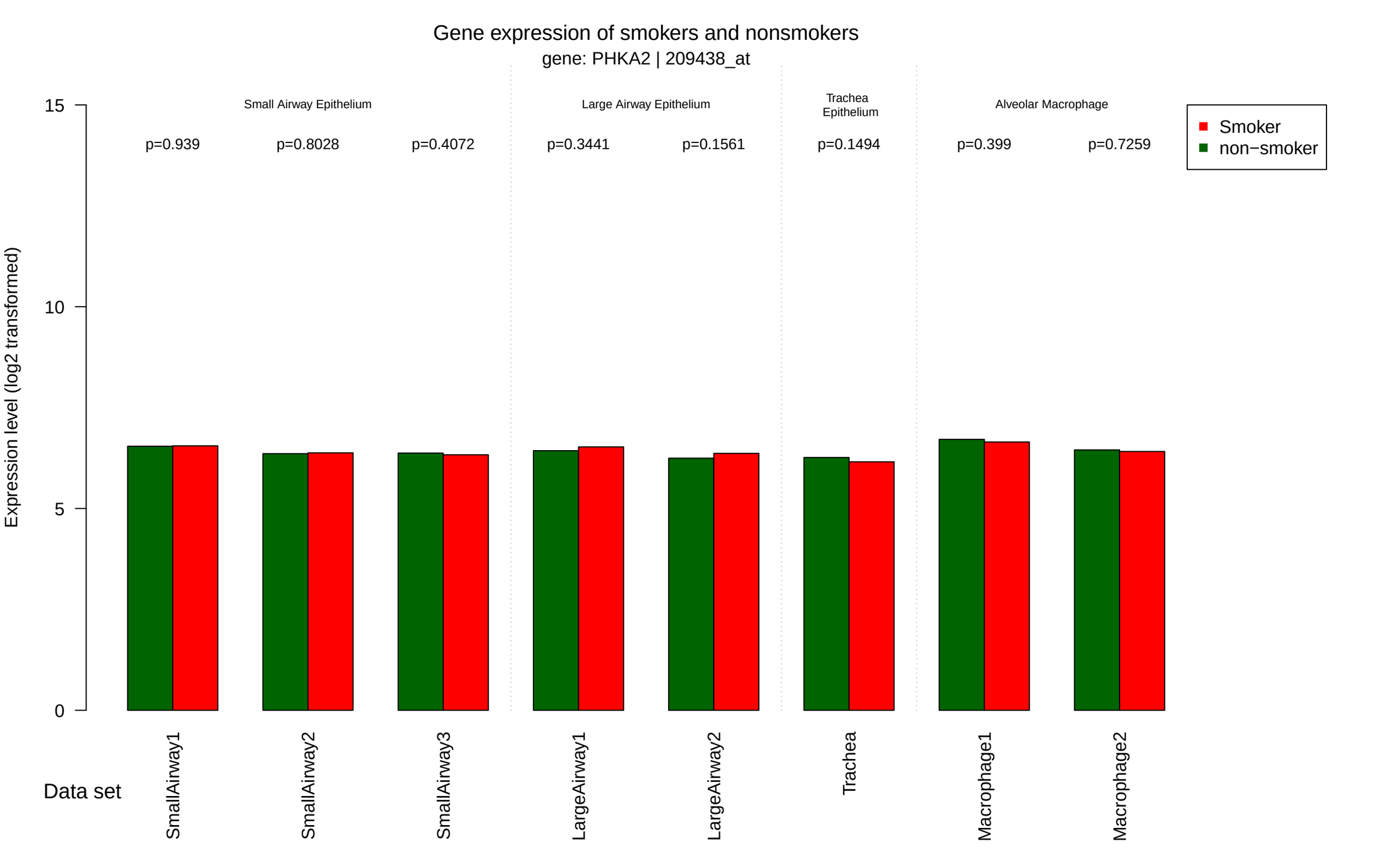
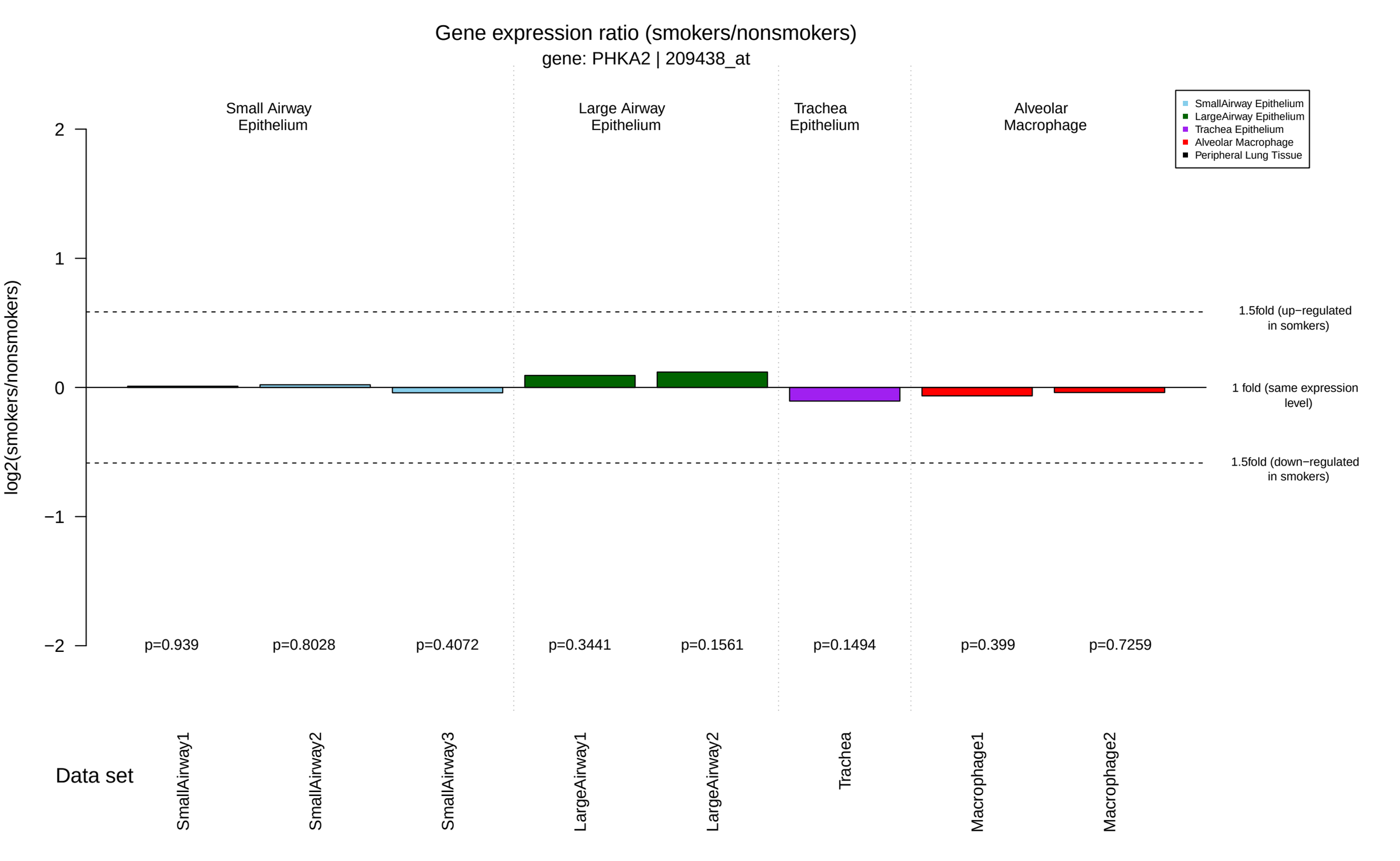
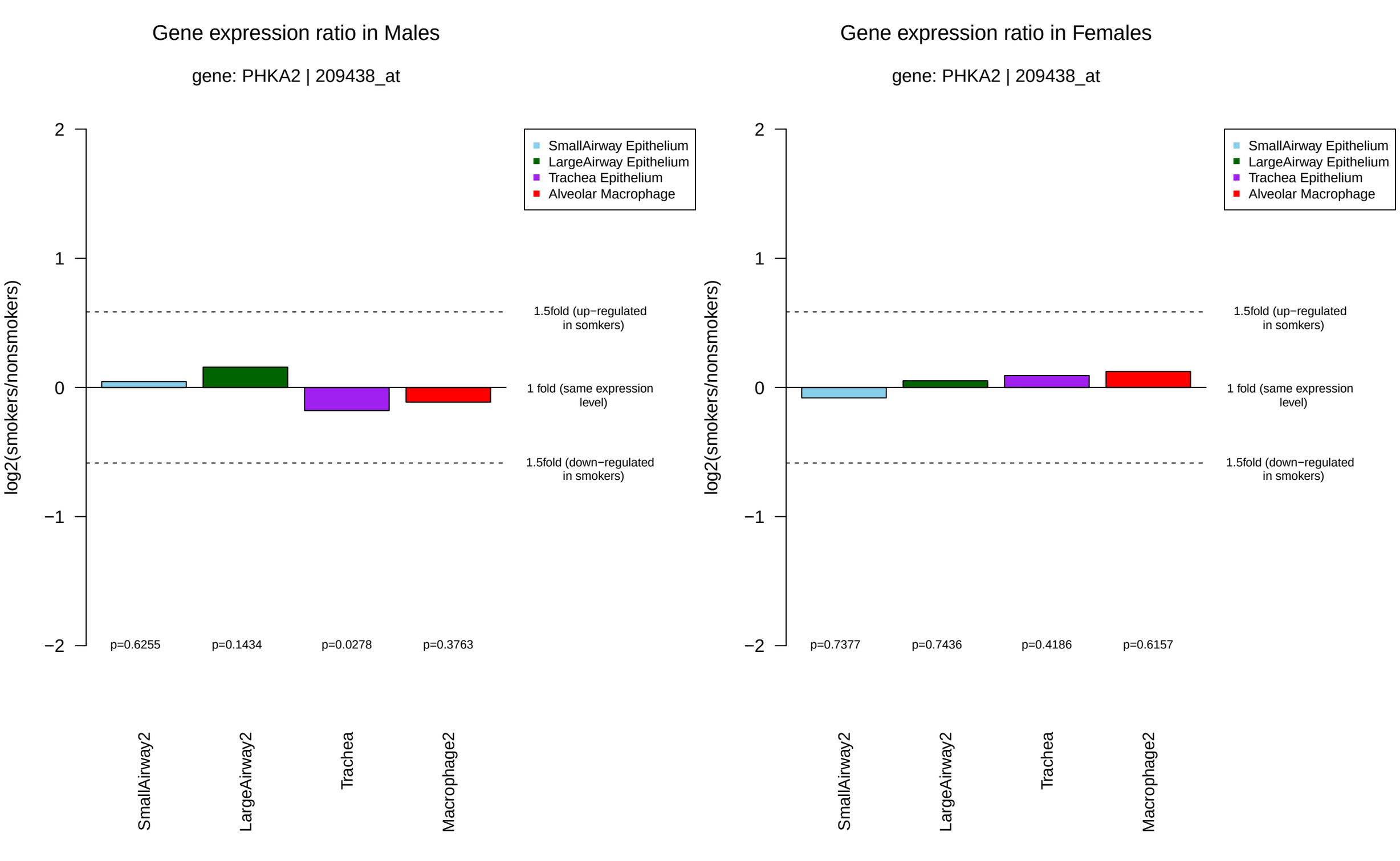
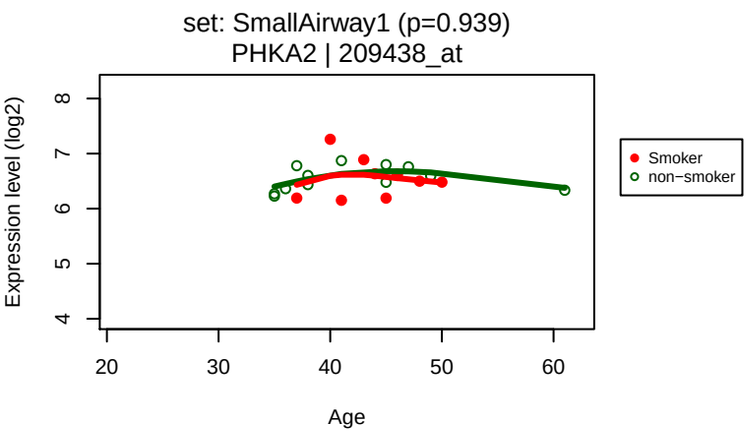
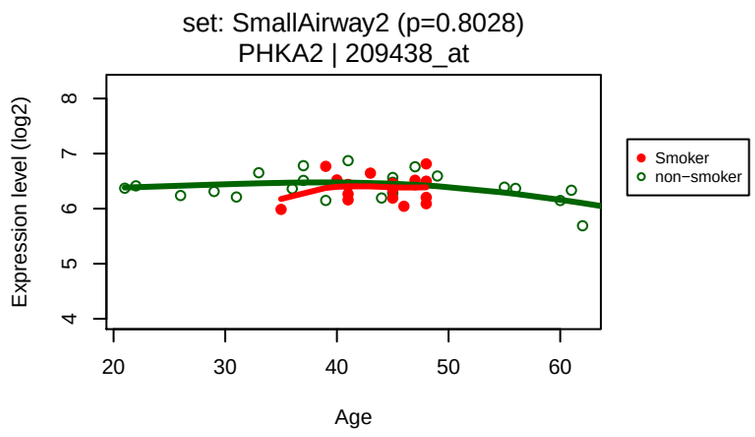
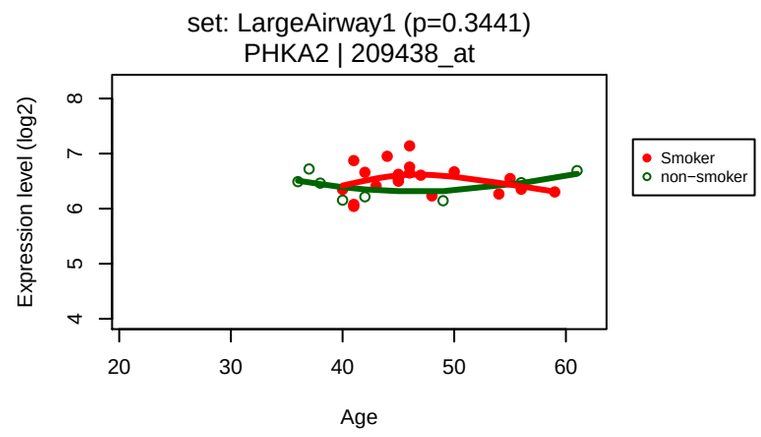

Gene expression of smokers and nonsmokers
gene: PHKA2 | 209438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.939
p=0.8028
p=0.4072
p=0.3441
p=0.1561
p=0.1494
p=0.399
p=0.7259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

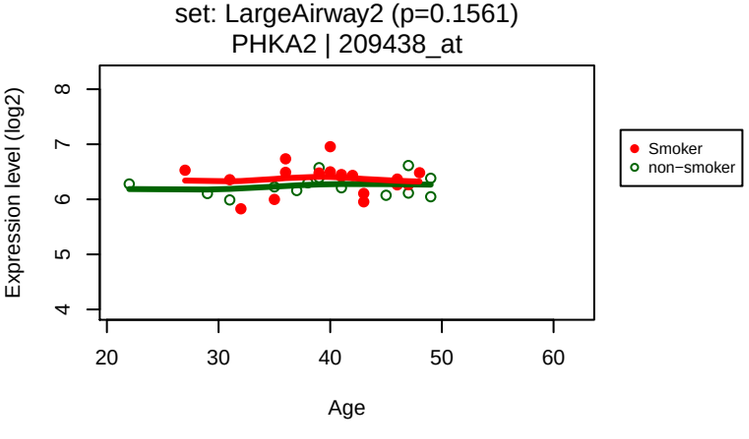

Gene expression ratio (smokers/nonsmokers)
gene: PHKA2 | 209438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.939
p=0.8028
p=0.4072
p=0.3441
p=0.1561
p=0.1494
p=0.399
p=0.7259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

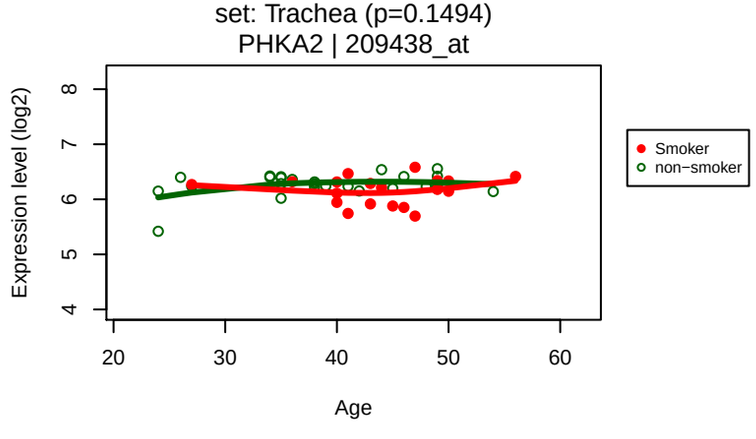

Gene expression ratio in Males
gene: PHKA2 | 209438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6255
p=0.1434
p=0.0278
p=0.3763
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PHKA2 | 209438_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7377
p=0.7436
p=0.4186
p=0.6157
SmallAirway2
LargeAirway2
Trachea
Macrophage2

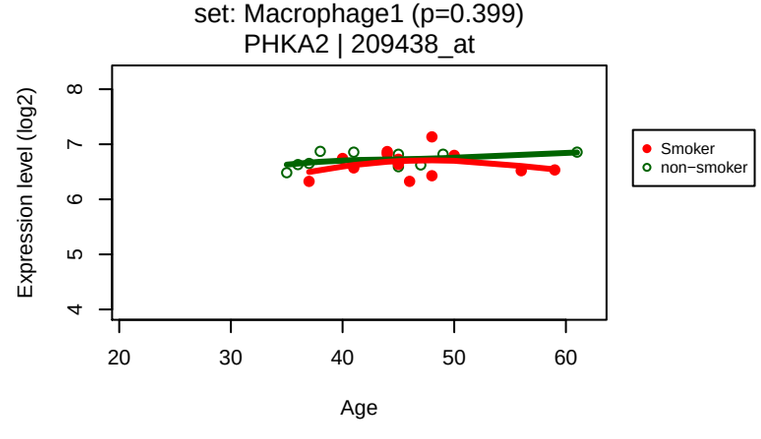

set: SmallAirway1 (p=0.939)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker

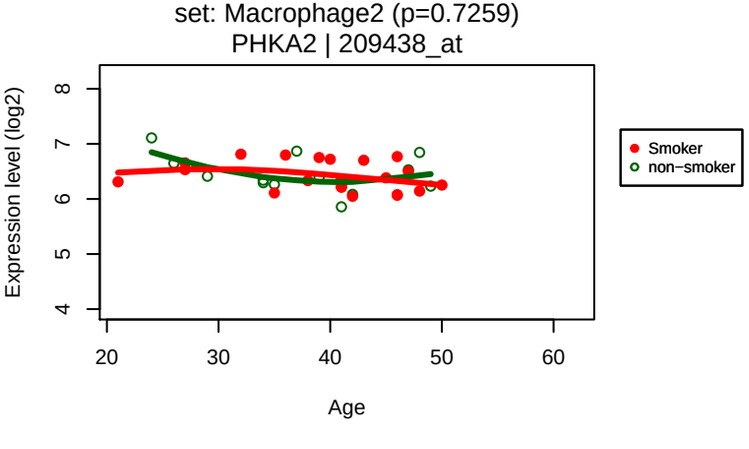

set: SmallAirway2 (p=0.8028)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker

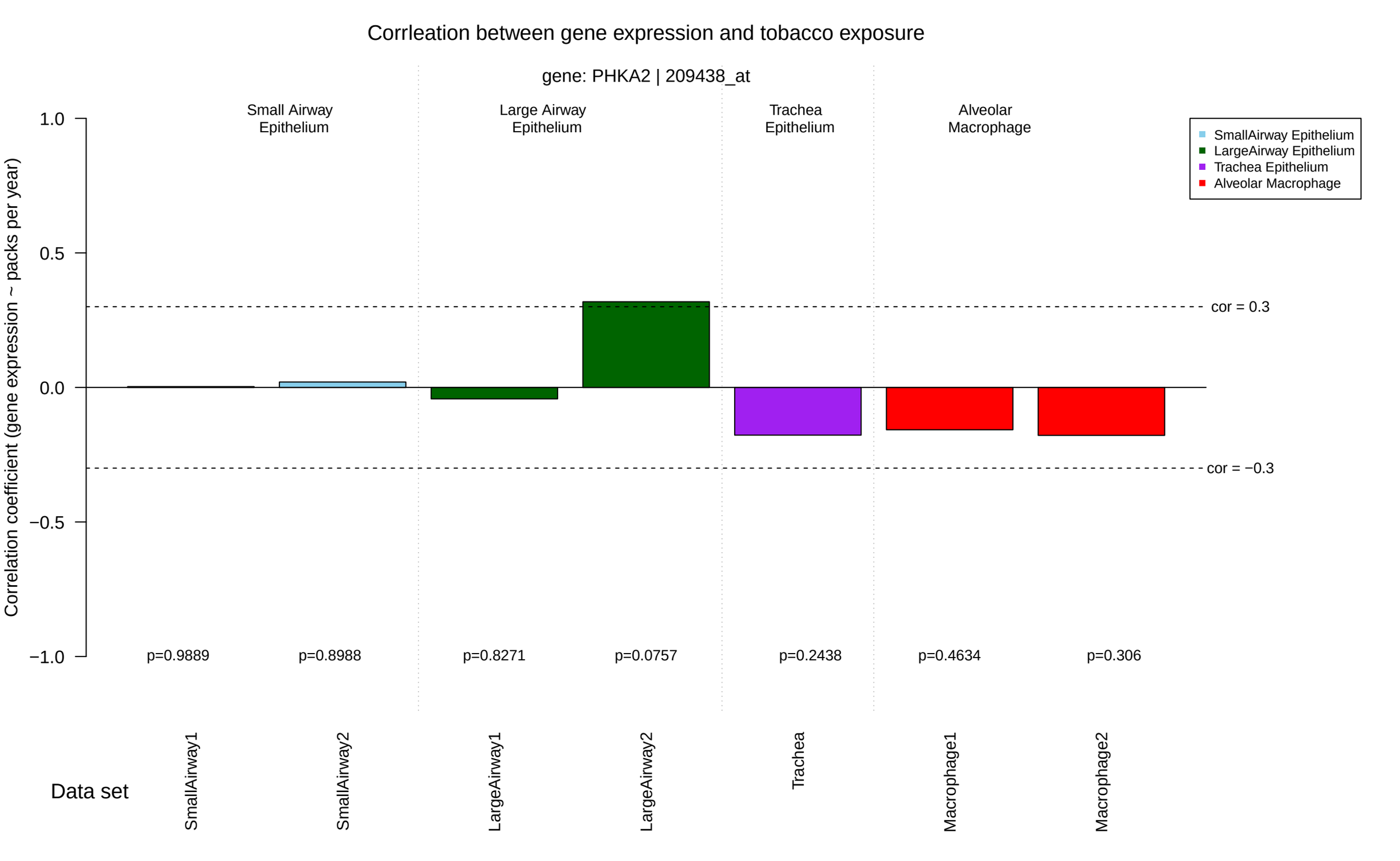

set: LargeAirway1 (p=0.3441)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1561)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1494)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.399)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7259)
PHKA2 | 209438_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PHKA2 | 209438_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9889
p=0.8988
p=0.8271
p=0.0757
p=0.2438
p=0.4634
p=0.306
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2